

TUTORIAL – 11

Python Automation Testing

Aim:

To analyze automation testing in Python for website navigation, forms, multi-page workflows, links, and regression testing scenarios.

Description:

Automation testing is a software testing technique in which software tools are used to execute test cases automatically.

In this practical, Python and Selenium WebDriver are used in **PyCharm IDE** to automate a web application. Students will perform browser navigation, form handling, links, multi-page workflow and regression testing.

The practical uses **Google Chrome + Selenium WebDriver + Python** only.

Problem Statement:

Develop and execute a Python Selenium automation script to test a web application by performing:

1. Website navigation
2. Login form handling
3. Web link interaction
4. Product selection
5. Multi-page workflow
6. Cart navigation
7. Checkout form handling
8. Validation testing
9. Regression testing

Implementation Details

Parameter Details

Parameter	Details
Project Name	Python Automation Testing
Application/Software	SauceDemo Web Application
Module Name	Navigation, Forms, Links and Multi-Page Workflow
SDLC Model	Agile
Programming Language	Python

IDE/Tool Used	PyCharm
Automation Tool	Selenium WebDriver
Browser	Google Chrome
Input/Test Data	Username, Password, Product and Checkout Details

Software / Environment Required

Students require only:

- PyCharm
- Python
- Google Chrome
- Selenium Python library
- Internet connection

Install Selenium in PyCharm Terminal:

```
pip install selenium
```

Procedure – How to Perform the Practical

Step 1 – Open PyCharm

Open PyCharm and create a new project:

SoftwareTesting

Step 2 – Create Python File

Create:

tutorial11.py

Step 3 – Install Selenium

Open:

PyCharm → Terminal

Run:

```
pip install selenium
```

Step 4 – Import Selenium

```
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC
```

Step 5 – Open Browser

```
driver = webdriver.Chrome()
driver.maximize_window()
```

Step 6 – Open Website

```
driver.get("https://www.saucedemo.com/")
```

Step 7 – Enter Login Details

Valid credentials:

Username: standard_user

Password: secret_sauce

Python code:

```
driver.find_element(By.ID, "user-name").send_keys("standard_user")
driver.find_element(By.ID, "password").send_keys("secret_sauce")
driver.find_element(By.ID, "login-button").click()
```

Step 8 – Validate Product Page

```
assert "inventory" in driver.current_url
```

Step 9 – Add Product

```
driver.find_element(
    By.ID,
    "add-to-cart-sauce-labs-backpack"
).click()
```

Step 10 – Open Cart

```
driver.find_element(
    By.CLASS_NAME,
    "shopping_cart_link"
```

```
).click()
```

Step 11 – Checkout

```
driver.find_element(By.ID, "checkout").click()
```

Step 12 – Fill Checkout Form

```
driver.find_element(By.ID, "first-name").send_keys("Student")  
driver.find_element(By.ID, "last-name").send_keys("Testing")  
driver.find_element(By.ID, "postal-code").send_keys("380001")
```

Click Continue:

```
driver.find_element(By.ID, "continue").click()
```

Step 13 – Validate Multi-Page Workflow

```
assert "checkout-step-two" in driver.current_url
```

Step 14 – Back and Forward Navigation

```
driver.back()  
driver.forward()
```

Step 15 – Close Browser

```
driver.quit()
```

Source Code

```
from selenium import webdriver  
from selenium.webdriver.common.by import By  
from selenium.webdriver.support.ui import WebDriverWait  
from selenium.webdriver.support import expected_conditions as EC  
  
driver = webdriver.Chrome()  
driver.maximize_window()
```

```
wait = WebDriverWait(driver, 10)
```

```
try:
```

```
    # 1. Open Website
```

```
    driver.get("https://www.saucedemo.com/")
```

```
    print("Website opened successfully")
```

```
    # 2. Login
```

```
    wait.until(
```

```
        EC.visibility_of_element_located(  
            (By.ID, "user-name")
```

```
        )
```

```
    ).send_keys("standard_user")
```

```
    driver.find_element(
```

```
        By.ID, "password"
```

```
    ).send_keys("secret_sauce")
```

```
    driver.find_element(
```

```
        By.ID, "login-button"
```

```
    ).click()
```

```
    # 3. Validate Products Page
```

```
    wait.until(
```

```
        EC.url_contains("inventory")
```

```
    )
```

```
    assert "inventory" in driver.current_url
```

```
    print("Login Test: PASS")
```

```
    # 4. Product Selection
```

```
    wait.until(
```

```
        EC.element_to_be_clickable(
```

```
            (By.ID, "add-to-cart-sauce-labs-backpack")
```

```
        )
```

```
    ).click()
```

```
    print("Product Selection: PASS")
```

```
    # 5. Open Cart
```

```
    driver.find_element(
```

```
        By.CLASS_NAME,
```

```
    "shopping_cart_link"
).click()

assert "cart" in driver.current_url

print("Cart Navigation: PASS")

# 6. Checkout
driver.find_element(
    By.ID,
    "checkout"
).click()

# 7. Checkout Form
wait.until(
    EC.visibility_of_element_located(
        (By.ID, "first-name")
    )
).send_keys("Student")

driver.find_element(
    By.ID, "last-name"
).send_keys("Testing")

driver.find_element(
    By.ID, "postal-code"
).send_keys("380001")

driver.find_element(
    By.ID, "continue"
).click()

# 8. Validate Multi-page Workflow
wait.until(
    EC.url_contains("checkout-step-two")
)

assert "checkout-step-two" in driver.current_url

print("Multi-Page Workflow: PASS")

# 9. Back and Forward
driver.back()
driver.forward()
```

```
print("Back/Forward Navigation: PASS")
```

```
finally:  
    driver.quit()
```

```
print("Tutorial 11 Completed Successfully")
```

SDLC–STLC Mapping Table

SDLC Phase	Corresponding STLC Phase	Testing Activity
Requirement Analysis	Requirement Analysis	Identify automation requirements
System Design	Test Planning	Prepare automation scenarios
Development	Test Case Development	Develop Selenium scripts
Testing	Test Execution	Execute Selenium scripts
Deployment	Test Closure	Verify test results
Maintenance	Regression Testing	Re-execute existing scenarios

Testing Activities Performed

Sr. No.	SDLC Phase	STLC Phase	Testing Activity Performed	Deliverable / Artifact
1	Requirement Analysis	Requirement Analysis	Identified automation requirements	Requirement List
2	System Design	Test Planning	Prepared automation scenarios	Test Plan
3	Development	Test Implementation	Developed Python Selenium script	Source Code

4	Testing	Test Execution	Executed automated tests	Test Execution Report
5	Maintenance	Regression Testing	Re-executed important scenarios	Regression Report

Deliverables of Each Phase

Sr. No.	SDLC Phase	Deliverable / Artifact	Purpose
1	Requirement Analysis	SRS	Defines software requirements
2	System Design	Test Plan	Defines testing approach
3	Development	Selenium Script	Automates test cases
4	Testing	Test Execution Report	Records results
5	Deployment	Test Summary Report	Summarizes testing
6	Maintenance	Regression Report	Verifies existing functionality

Roles & Responsibilities

Sr. No.	Role	Responsibility	Phase Involved
1	Project Manager	Manages project activities	All Phases
2	Business Analyst	Defines requirements	Requirement
3	System Analyst	Analyzes system	Requirement/Design
4	Software Developer	Develops software	Development
5	Software Tester / QA Engineer	Performs testing	Testing
6	Database Administrator	Manages database	Development/Testing
7	Client / End User	Provides feedback	Testing/Deployment

Test Case Execution

TC ID	Test Scenario	Expected Result	Actual Result	Status
TC-11-01	Open website	Website opens	Website opened	Pass
TC-11-02	Valid Login	User logs in	User logged in	Pass
TC-11-03	Product Selection	Product added	Product added	Pass
TC-11-04	Open Cart	Cart opens	Cart opened	Pass
TC-11-05	Checkout Form	Data accepted	Data accepted	Pass
TC-11-06	Multi-page Validation	Correct page opens	Correct page opened	Pass

Observation

Sr. No.	Observation	Remarks
1	Chrome launched through Selenium	Successful
2	Login form automated	Successful
3	Product added to cart	Successful
4	Checkout form completed	Successful
5	Multi-page workflow validated	Successful

Output

Students should capture screenshots of:

1. PyCharm project
2. Source code
3. Login page
4. Product page
5. Cart page
6. Checkout page
7. PyCharm console output

Questionnaire

Q1. Which Python library is commonly used for web automation testing?

- A) NumPy
- B) Pandas
- C) Selenium
- D) Matplotlib

Answer: C) Selenium

Q2. Selenium WebDriver is mainly used for?

- A) Database Management
- B) Browser Automation
- C) Network Configuration
- D) Operating System Testing

Answer: B) Browser Automation

Q3. Which command installs Selenium in Python?

- A) install selenium
- B) python selenium
- C) pip install selenium
- D) selenium setup

Answer: C) pip install selenium

Q4. Regression Testing is performed to ensure that:

- A) New functionality is developed
- B) Existing functionality remains unaffected after changes
- C) Database is created
- D) Source code is deleted

Answer: B) Existing functionality remains unaffected after changes

Q5. Which Selenium component is used for browser automation?

- A) Selenium Grid
- B) Selenium IDE
- C) Selenium WebDriver
- D) Selenium Server

Answer: C) Selenium WebDriver